

NFC-Based Attendance System

When I launch my **Smart Attendance Logger**, I initiate a coordinated system composed of both hardware and software components designed to streamline attendance tracking. At the hardware level, an **Arduino Uno** communicates with a **PN532 NFC reader** and an **HM-10 Bluetooth Low Energy (BLE) module**. When a student taps their **Visa or Apple Pay-compatible card**, the system follows the **EMV protocol**, extracting both the **Primary Account Number (PAN)** and the **expiry date**. These two pieces of data are concatenated into a unique identifier (UID), which is transmitted over both **USB serial** and **BLE** using a simple, efficient `println(uid)` call.

On the software side, a Python-based **transport layer** initializes before the user interface appears. This layer scans for available **COM ports** using `pyserial`, and simultaneously searches for **advertising BLE peripherals** such as the HM-10 using the `bleak` library. Once a valid connection is established—whether wired or wireless—the corresponding channel begins streaming data into a unified, thread-safe queue. This abstraction ensures that the source of the UID (USB or BLE) is transparent to the rest of the application.

The **Tkinter GUI** serves as the user-facing interface. Upon startup, the application presents a clean, modern layout with a cyan-colored status banner that provides real-time feedback on connection progress. Phrases such as “*Scanning...*”, “*Connecting...*”, and “*Subscribed — streaming*” appear sequentially within the same label, maintaining visual consistency. Once a BLE connection is fully established, the status transitions to **blue**, and shortly thereafter turns **green**, displaying “*Ready to scan...*” to indicate that the system is operational.

Functionally, the system integrates with an existing **Excel workbook (attendance.xlsx)** using the `openpyxl` library, while preserving embedded **VBA macros**. At runtime, the application locates the current date in the attendance sheet header (row 9), and matches scanned UIDs against entries starting from row 10. If a match is found, the student is marked present for the day. If no match exists, a dialog prompts the administrator to input the student’s name and phone number, which are then stored in the Excel sheet alongside the UID.

Additionally, the interface includes an optional checkbox labeled “*Do you have a second Device?*”. When selected, this activates a background thread that waits for a second UID scan—useful for scenarios where a secondary identifier is required for the same student.

In summary, this project enables **rapid, contactless attendance logging** using both wired and wireless communication, with **Excel integration** for compatibility with institutional recordkeeping. Its user interface provides clear visual cues throughout the process, and

the dual-transport design ensures flexibility and reliability. This solution significantly reduces administrative overhead, improves accuracy, and supports a modern, tech-enabled classroom experience.